pair in a semiconductor is only a few electronvolts, while energy to create an ion pair in air is an order of magnitude larger at ~ 34 eV

Advantages of semiconductor dosimeters over air-filled ionization chambers are as follows: (i) higher sensitivity per unit cavity volume, (ii) higher spatial resolution, (iii) higher mechanical stability and strength. Since diamond dosimeter has several advantages over silicon diode dosimeter, one can conclude that in many respects diamond dosimeter offers several advantages of ionization chamber; however, it is not ready yet for absolute dosimetry.

17.4 Film Radiation Dosimetry

Medical film radiation dosimetry falls into the group of relative radiation dosimetry techniques and comprises two major film categories: (i) radiographic film and (ii) radiochromic film.

Both types of dosimetric film serve the same purpose and give similar results; however, the use of radiographic film in dosimetry is waning while radiochromic film is gaining in importance. The main reason for this shift is the significant difference in film processing required by the two film types; processing of radiographic film requires access to sophisticated developers and fixers, while radiochromic film is self-processing and requires no additional film processing equipment.

In the past film developers were readily available in clinics and hospitals that were using radiographic films for diagnostic procedures and they could also be used for radiation film dosimetry. During the past decade, however, the use of analog radiographic film for diagnostic imaging has been discontinued in the developed world in favor of digital computerized radiographic techniques. This shift caused the disappearance of film developers from hospitals and clinics and provided a strong economic incentive for the use of radiochromic film for medical film radiation dosimetry.

17.4.1 Absorbance and Optical Density

Film-based radiation dosimetry depends strongly on transmission of light through processed film, be it radiographic or radiochromic. The transmission is affected by film opacity that can be measured in terms of optical density *OD* with devices, such as film densitometer, laser densitometer, automatic film scanner, and spectrophotometer. Similarly to absorbance A_λ of a Fricke solution, discussed in Sect. 16.2.9 and measured with a spectrophotometer, optical density *OD* of film depends on absorbed dose and is defined as

$$OD = \log \frac{I_\mathrm{o}}{I} \qquad \text{or} \qquad I = I_\mathrm{o} 10^{-OD} = I_\mathrm{o} e^{-(2.302)OD}, \tag{17.20}$$

where

I_0 is the initial light intensity emitted from a light source, such as a light emitting diode (LED), laser, or white light fluorescent source.

I is the light intensity transmitted through film and measured by a sensor such as a photodiode, photomultiplier tube (PMT), or linear charge coupled device (CCD).

For example, if a processed film transmits 10% of the incident light beam, then $I/I_0 = 0.1$ or $I_0/I = 10$, and the optical density of the film is $OD = \log 10 = 1$. Similarly, an optical film density $OD = 2$ corresponds to film transmission fraction $I/I_0 = \exp(-2.302\times 2) = 0.01$.

In clinical practice, optical densities range from less than 1 to over 3. The advantage of the logarithmic definition of optical density given in (17.20) is that it makes optical densities additive. Optical density of two layers of film or two exposures of the same film are additive and the total optical density is the sum of two densities ($OD_{\text{tot}} = OD_1 + OD_2$). A densitometer that has been calibrated by a standard strip of film containing regions of known optical density can provide a direct reading of optical density.

A graph of optical density OD against absorbed dose of a typical dosimetric film is referred to as the sensitometric curve and is also called characteristic curve of film or the H&D sensitometric curve in honor of Ferdinand Hurter and Vero C. Driffield who were the first to investigate the relationship. A typical H&D characteristic curve for a dosimetric film is shown in Fig. 17.17. The curve has four regions:

(i) Base + fog at very low doses.
(ii) Toe in transition region between base and the linear region.
(iii) Linear region at intermediate doses.
(iv) Shoulder in the transition from the linear region to saturation.
(v) Saturation at high doses.

Note: The dosimetric quantity of interest is usually the net optical density (netOD) or OD_{net} that is defined as the measured optical density OD_{meas} less the base optical density OD_{base}, or

$$OD_{\text{net}} = OD_{\text{meas}} - OD_{\text{base}}. \tag{17.21}$$

General characteristics of radiographic and radiochromic film dosimetry:

1. While film is not useful for absolute dosimetry, film dosimetry possesses several attributes that make it very useful and practical in special areas of radiation dosimetry as well as in acceptance testing, commissioning, and quality assurance of high technology equipment used in radiation medicine.
2. Film has an excellent spatial resolution and is often used to measure 2D dose distribution maps, especially in regions of high dose gradient and regions of dynamic dose.
3. Film is especially useful for gaining qualitative data that provide fast general information on radiation dose distribution.

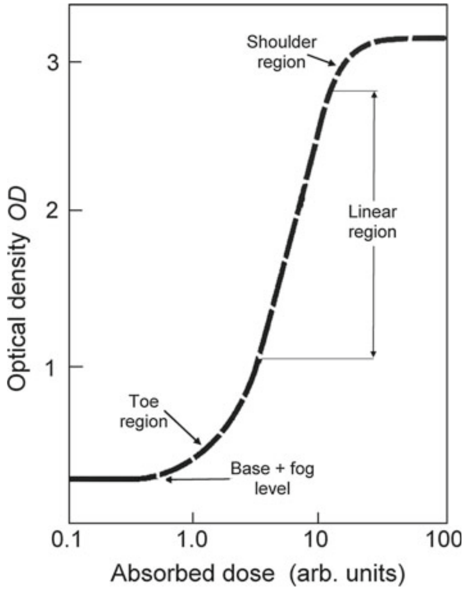

Fig. 17.17 A typical H&D characteristic curve for a dosimetric film showing five regions: (i) base + fog, (ii) toe, (iii) linear region, (iv) shoulder, and (v) saturation

17.4.2 Radiographic Film Dosimetry

Radiographic film has been in use for x-ray detection since Wilhelm K. Röntgen discovered x-rays in 1895. The discovery of x-rays was translated into hospitals and clinics around the world for medical purposes within a few weeks of their discovery and x-ray images produced with steady improvements in film technology have formed the backbone of radiology for more than a century.

Composition of Radiographic Film

Modern radiographic film is typically about 0.25 mm thick and comprises several layers, as shown in Fig. 17.18(a). The radiographic film layers and their functions are as follows:

(i) Film base – made of polyester resin that provides support for active layers of film.
(ii) Adhesive layer – binds emulsion to film base.
(iii) Emulsion – active layer of film; comprises gelatin and uniformly distributed small silver halide crystals.
(iv) Protective coating – protects the active film layer.

Note: Layers listed above form single sided films; to increase sensitivity films are double sided containing the same layers on each side of the film base (in same order starting from the film base).

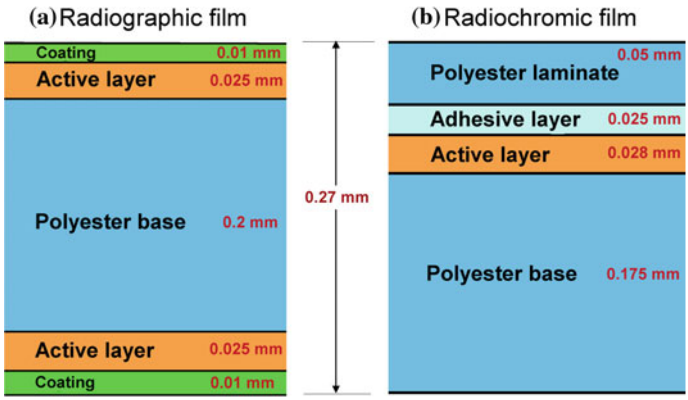

Fig. 17.18 Transverse section through two typical film types used in film radiation dosimetry: **a** Double-sided radiographic film with silver halide based active layer and **b** Single sided EBT2 radiochromic film with monomeric di-acetylene based active layer

Emulsion forms the active film layer of which silver halide crystals (~95% silver bromide AgBr and ~5% silver iodide AgI) form the most important component. The silver halide crystals (also called grains) are ionic crystals composed of silver ions Ag^+ and halogen ions Br^- and I^-. The microscopic crystals, arranged into a lattice structure, are not perfect nor are they chemically pure. Rather, they contain imperfections called sensitivity specks that are either naturally occurring (edge dislocations) or are introduced into the crystals through doping (impurities) to increase sensitivity. It is estimated that each crystal grain contains about 10^{10} atoms and is about 1 to 2 μm in diameter.

Another important imperfection in the AgBr crystal is the so-called Frenkel defect that is characterized by silver ions Ag^+ located interstitially in the lattice with their corresponding negatively charged silver ion vacancies. The interstitial Ag^+ ion is extremely mobile through the Ag^+Br^- ionic lattice and serves as the basis for forming the latent image during the exposure of film to visible light or ionizing radiation.

The photographic effect as well as the utility of radiographic film in radiation dosimetry arises from the lattice defects that occur in imperfect AgBr crystals. These lattice defects serve as shallow traps for free electrons that are located in the energy band below the conduction band in the energy gap separating the conduction band from the valence band (see Sect. 17.2.1).

Formation of Latent Image in Radiographic Film

During film exposure, ionizing radiation releases energetic free electrons (photoelectrons, Compton electrons, etc.) in the film and these electrons through ionization of atoms and ions of the film components release free electrons into the conduction band of the AgBr crystals. Some of the free electrons, while migrating through silver halide crystals get close to a sensitivity speck and may get trapped into a shallow electron trap.

An electron trapped in an electron trap (sensitivity speck) attracts a nearby interstitial silver ion Ag^+ from a Frenkel defect and reduces it to neutral Ag^0 atom ($Ag^+ + e^- \rightarrow Ag^0$). If the reduction process is repeated at least another 3 times on a given crystal, a latent image that can be made visible through film processing is formed. The extent to which a given area of film is affected by ionizing radiation depends on the number of particles (photons or charged particles) incident upon that area.

After exposure to ionizing radiation, radiographic film contains a latent image that is not visible before the film undergoes appropriate chemical processing that consists of two steps: (i) film development and (ii) film fixation.

Radiographic Film Processing

During development the silver bromide crystals that have been sufficiently affected during exposure to radiation (typically 4 hits or more) are converted into opaque black silver specks with all Ag^+ ions reduced to Ag^0 atoms and all Br^- ions removed. The crystals that have not been affected during exposure remain in their original state in the emulsion. A chemical suitable as film developer must, on the one hand, be able to donate rapidly electrons to crystals containing a latent image and, on the other hand, be much slower in donating electrons to crystals that do not contain a latent image. The duration of the development process must be carefully timed to prevent eventual complete reduction of all silver ions to atomic silver irrespective of whether or not they contain latent image.

During fixation the unaffected AgBr crystals in the emulsion are dissolved leaving behind the record of radiation exposure in the form of a corresponding pattern of opaque black silver specks distributed over the gelatin. The metallic silver is unaffected by the fixer but causes darkening of the developed film and the degree of darkening depends on radiation exposure of the film that can be measured with a film densitometer.

17.4.3 General Characteristics of Radiographic Film

Radiographic film has a long and illustrious history dating back to the discovery of x-rays in 1895. Since then it was used both in the clinic for imaging as well as in radiation dosimetry as detector of ionizing radiation.

Dosimetry based on radiographic film consists of the following four steps:

1. Film manufacture – crystals of silver halide (mainly AgBr) of appropriate size and containing a desirable concentration of lattice defects (sensitivity specks) are suspended uniformly in gelatin to produce film emulsion. The emulsion is bound onto polyester fim base and coated with a protective layer.
2. Exposure to ionizing radiation (photons, electrons, protons) to create a latent image in the active film layer.
3. Radiographic film processing consists of three stages: (a) Development of exposed film to convert the latent image to atomic silver Ag^0; (b) Fixation of developed film to dissolve and remove the AgBr remaining on the developed film and to

harden the gelatin; (c) Washing the fixated film in pure water and air-drying the fixated film.

4. Measurement of optical film density to determine absorbed dose based on measurement of the film H&D sensitometric curve.

When used in radiation dosimetry, radiographic film is characterized by several advantages and disadvantages in comparison with other dosimetric techniques. The significant advantages of radiographic film are:

- Very high spatial resolution limited by the physical size of the light source of the densitometer rather than by the film itself.
- Permanent storage of processed image and dosimetric information allowing multiple non-destructive readouts.
- Convenient physical shape of the dosimeter cavity (small thickness and arbitrarily large area) and flexibility allowing easy use in radiation field mapping and general quality assurance measurements of photon and electron beams.
- Until recently, widespread commercial availability of a variety of radiographic film types and shapes. The advent of digital radiography has significantly curtailed the manufacturing of radiographic films and severely limited access to radiographic film developers, so that this feature can no longer be considered an advantage of radiographic films.

Some of the notable disadvantages of radiographic film are:

- Radiographic film can only be used as a relative dosimetry technique, since the response of a radiographic film dosimetry system must be calibrated in a known radiation field before use for measurement of absorbed dose.
- Radiographic film is not tissue equivalent because it contains silver bromide that increases its effective atomic number above that of water and tissue.
- Radiographic film exhibits strong energy dependence of film response for photons with energy below $\sim$100 keV as a result of the high atomic number of silver ($Z = 47$) and predominance of photoelectric effect at low photon energies (see Sect. 8.2.3, Fig. 8.5). For example, the relative response per unit of x-ray exposure normalized to cobalt-60 γ-rays for a typical radiographic film is about 20 times higher for 50 keV x-rays and the ratio of mass energy absorption coefficients for typical radiographic film emulsion to water at 50 keV x-rays is $\sim$100 to 1.
- Radiographic film processing involves wet chemical development and fixation that both require strong quality control and meticulous equipment maintenance to obtain results that are in line with standards of radiation dosimetry.
- Radiographic film is sensitive to visible light, so any work involving open film must be carried out in a dark room.

17.4.4 *Radiochromic Film Dosimetry*

Radiochromic dosimetry is a chemical dosimetry technique based on radiochromic reactions defined as reactions that are: (i) triggered by ionizing radiation and (ii) result in a color change in the radiation sensitive medium. The radiochromic dosimetry has been developed during 1960s for high dose measurements in the kGy range for industrial applications of ionizing radiation.

As the sensitivity of radiochromic techniques was improved to the level of medical dosimetry, it became apparent that radiochromic dosimeters could be fashioned into the format used for radiographic film dosimetry. The goal was to develop radiochromic film systems with characteristics that match, or improve upon, those attributed to radiographic film. However, unlike radiographic films that are based on silver halides, radiochromic films would be based on radiochromic reactions producing a color change in the irradiated detector that is proportional to the dose absorbed in the detector. Hence, measured color change in radiochromic film exposed to ionizing radiation would be used for relative radiation dosimetry.

Radiochromic Film Based on Polymerization of Di-Acetylene

In the past, many organic materials and several chemical processes were used in radiochromic radiation detectors; however, none of these techniques were sensitive enough for medical applications. This changed during 1980s with the development of a new radiochromic detector in the format of radiographic film. The active radiation sensitive medium of the new radiochromic film is based on di-acetylene (C_4H_2), a highly unsaturated hydrocarbon monomer that contains three single bonds and two triple bonds ($H - C \equiv C - C \equiv C - H$). The film consists of thin microcrystalline monomeric di-acetylene emulsion coated on a flexible polyester film base. A schematic diagram of a typical radiochromic film is given in Fig. 17.18b for comparison with a typical radiographic film shown in Fig. 17.18a. Pioneers in the development of medical radiochromic film dosimetry are William McLaughlin, David F. Lewis, and Christopher G. Soares.

The radiochromic film is translucent before irradiation. Under exposure to ionizing radiation the di-acetylene monomers undergo progressive chain-growth photo-polymerization. This results in formation of colored polymer chains that grow in length and the film turns to a shade of blue that becomes darker as the absorbed dose increases.

According to the IUPAC (International Union of Pure and Applied Chemistry), polymerization is the process of converting a monomer or a mixture of monomers into a polymer and photo-polymerization is chain-growth polymerization initiated by visible light, ultraviolet light, or ionizing radiation.

It is notable that no physical, chemical or thermal processing is required to develop or fix the radiochromic film image – radiochromic film is self-developing; however, after exposure to ionizing radiation, polymerization slowly continues for about 24 hours post-irradiation and then color stability sets in.

Several companies are involved in production of radiochromic films for various applications. The majority of these films require doses that are much higher than those used in clinical work and the films are then used for commercial purpose such as food irradiation, medical equipment sterilization, and waste management. The best-known medical radiochromic film is the GAFchromic EBT2 film produced by International Specialty Products, Wayne, NJ, USA.

Absorption Spectrum of Radiochromic Film

The dosimetric signal in medical radiochromic film dosimetry is obtained from the color change or film darkening that is proportional to the dose absorbed in the film active layer. The amount of light that is transmitted through (or absorbed in) the film during the readout procedure depends on the dose of ionizing radiation and the wavelength λ of the light used in the readout device.

Each type of radiochromic film has its own absorbance spectrum and an example of such spectrum is given in Fig. 17.19 for the EBT2 radiochromic film in the wavelength region from 350 nm to 800 nm. Two absorption curves are shown: one for unexposed film and one for film exposed to 200 cGy of cobalt-60 gamma rays. The curves show three absorbance peaks: two dosimetric peaks (Peak 2 at 583 nm and Peak 3 at 635 nm) and one broad non-dosimetric peak (labeled Peak 1) at about 420 nm. The non-dosimetric Peak 1 is attributed to the addition of yellow dye to the active film layer for inhibition of film sensitivity to visible and ultraviolet light.

Measurement of the absorption spectrum of a given radiochromic film type is important, since it allows optimization of the measured film sensitivity during the film analysis. Generally, the light source of the film readout device is selected such that it matches as close as possible the absorption maximum of a dosimetry peak of the given radiochromic film. For example, a helium-neon laser with emission of red light at 633 nm has been found a good match for the dosimetry peak at 635 nm of the EBT2 film.

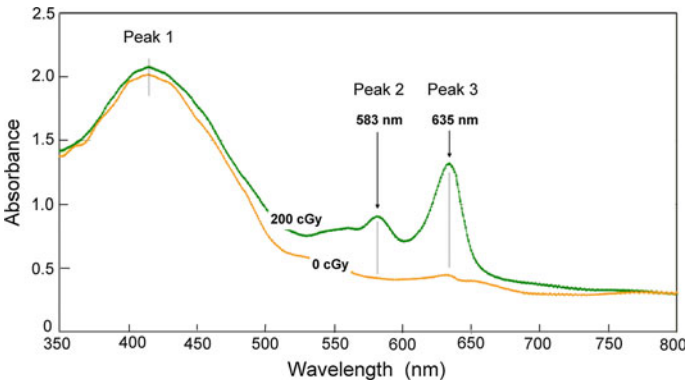

Fig. 17.19 Absorbance spectrum (adapted from David F. Lewis, ISP Corporation) for GAFchromic^{TM} EBT2 film in the wavelength range from 350 nm to 800 nm for: (i) unexposed film and (ii) film exposed to 200 cGy of cobalt-60 gamma rays. Exposed film clearly shows two radiation induced absorbance peaks (dosimetric peaks), peak 2 at 583 nm and peak 3 at 635 nm. The non-dosimetric absorbance peak at ~ 420 nm (peak 1) is attributed to the addition of yellow dye into the active layer of the radiochromic film. The yellow dye significantly decreases the film sensitivity to ambient fluorescent light; however, the absorbance peak is not of much value for dosimetry

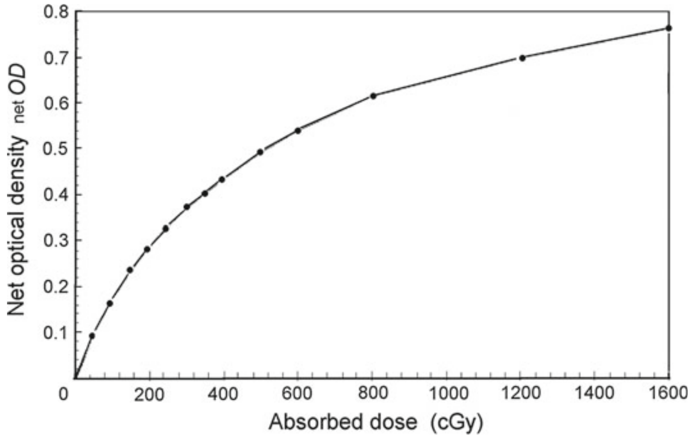

Fig. 17.20 Typical example of an H&D sensitometric curve of EBT2 radiochromic film. The plot "net*OD*" against dose would serve as a calibration curve in radiochromic film dosimetry using the same radiochromic film batch, same measuring equipment, and same radiation source

Calibration of Radiochromic Film

Since medical film dosimetry is a relative dosimetry technique, it is imperative that the radiochromic film technique be calibrated before use for measurement of unknown doses. The calibration procedure consists of the determination of the H&D sensitometric relationship for the radiochromic film and must encompass all components of the measuring system. Moreover, the calibration must be carried out on the specific radiation source that will be used for measuring unknown doses. The measured H&D calibration curve should encompass the dose region of future interest; extrapolation out of the calibration region is not acceptable.

A typical H&D sensitometric calibration curve for EBT2 radiochromic film is shown in Fig. 17.20 with a plot of net*OD* against absorbed dose. The plot would serve as the calibration curve for the specific film batch, measuring equipment, and radiation source. The measured net*OD* of an unknown dose can be used in conjunction with the calibration curve to determine the actual dose delivered to the radiochromic film.

17.4.5 Comparison Between Radiographic and Radiochromic Radiation Dosimetry

Radiographic film has for over a century been used in medicine for clinical imaging as well as radiation dosimetry. Its use in both areas has been waning during the past decade as a result of the advent of digital radiography techniques in clinical x-ray imaging and advances made in radiochromic film technology in terms of greater film sensitivity and decreased cost for use in film dosimetry. Clinical use of radiographic films has essentially disappeared in the developed world in favor of digital imaging

Table 17.4 Comparison of characteristics of radiographic film dosimetry and radiochromic film dosimetry

Characteristics	Radiographic film	Radiochromic film
Radiation sensitive medium	Silver halide AgBr (90%) + AgI (10%)	Di-acetylene C_4H_2 monomer
Radiation induced process	Reduction $Ag^+ + e^- \rightarrow Ag^0$	Polymerization of di-acetylene monomer into polymer
Post-irradiation film processing	Dark room Film processing	**NO**
Darkroom required	YES	**NO**
Film spatial resolution	High	High
Water equivalence	NO	$\sim$ **YES**
Signal low-energy dependence	Severe	**Slight**
Dose rate dependence	NO	NO
Visible light sensitivity of unexposed film	YES	**NO**
Ultraviolet light sensitivity of unexposed film	YES	YES
Allows direct water immersion during irradiation	NO	**YES**

techniques. This has caused the disappearance of darkrooms and well-maintained film processors from hospitals and clinics and severely curtailed the availability of radiographic film dosimetry in radiology and radiotherapy departments.

It is simply economically prohibitive to maintain a film processor in a standard medical physics department just for occasional use in film dosimetry, especially if another technique, such as radiochromic film dosimetry that, in addition, features a few other advantages, requires no film processing. Evidently, the significant physical advantages of radiochromic films over radiographic films, combined with economic considerations, have given medical radiochromic film dosimetry a clear advantage over radiographic film dosimetry. Table 17.4 provides a comparison of various pertinent characteristics of traditional radiographic film dosimetry and the novel polymeric radiochromic film dosimetry.

A brief comparison of several important dosimetric parameters for the two known film radiation dosimetry techniques yields the following results:

1. *Effective atomic number* Z_{eff}: In contrast to radiographic film that relies on dosimetric signal produced in silver halides embedded in active film layer, all medical radiochromic films have effective atomic number Z_{eff} close to and slightly lower than that of water, since their radiation sensitive material is organic and contains mainly hydrogen and carbon. For example, Z_{eff} of the EBT2 radiochromic film is $\sim$6.8 compared to $\sim$7.5 for water. This means that the active sensor medium has a mass collision stopping power similar to that of water and mass energy absorption coefficient for photons of energy $h\nu$ exceeding 100 keV also similar to that of water.

2. *Water equivalency*. Based on Z_{eff} one may conclude that radiochromic film is almost water equivalent and its response is essentially independent of energy for photons above 100 keV. For photon energies below 100 keV the situation is not as clear: for some radiochromic films the response is actually up to 40% below that at 1 MeV and for other films it is slightly above. However, the low energy response of radiochromic films is far less pronounced than that of radiographic films where the over-response at low energies exceeds the response at 1 MeV by a factor of one order of magnitude or more.
3. *Dose rate dependence*. Similarly to radiographic film, radiochromic films do not exhibit a dependence on dose rate.
4. *Environmental factors*, such as ambient temperature and humidity during irradiation and readout, affect the radiochromic film response and have to be strictly controlled for best results.
5. *Sensitivity to visible and ultraviolet photons*. Radiochromic film is **not** sensitive to photons above 300 nm but is affected by ultraviolet light. This means that exposure to incandescent lighting is fine; however, ambient fluorescent light and sunlight must be avoided so as not to trigger photo-polymerization with the ultraviolet light that would be of non-dosimetric origin.
6. *Post-irradiation processing*. In contrast to radiographic film, the radiochromic film requires no post-irradiation processing, no chemicals, and no darkroom. Radiochromic film is self-developing through the process of radiation-induced polymerization of di-acetylene monomers that starts within a fraction of a second after the start of irradiation and stabilizes within about 24 hours post-irradiation.
7. *Cutting and immersion in water*. Radiochromic film can be cut to desired shape at room temperature in ambient light in preparation for use in anthropomorphic phantoms. It can also be safely immersed in water.
8. *Polarization effects*. New radiochromic films exhibit film orientation dependent polarization effects. Measured optical density may vary as the polarization plane of the analyzing light changes with film orientation. The effect is most severe when laser beam is used for readout. Film mounting convention on the readout device should be consistent with calibration procedure.

17.4.6 Main Characteristics of Radiochromic Film

Radiochromic radiation dosimetry is a relative chemical dosimetry technique that relies on radiation induced radiochromic reactions for measurement of absorbed dose. These reactions produce in the radiochromic film (radiation dosimeter cavity) a color change that is proportional to the dose absorbed in the dosimeter and can be measured by determining the optical density of the radiochromic film with a film densitometer.

Radiochromic film dosimetry has been known for several decades but has only during the past two decades become a serious competitor to radiographic film dosimetry, largely because of significant improvements in radiochromic film technology in terms of increased film sensitivity, improved understanding of the chemistry behind the radiochromic process, and decreased film production cost. Radiochromic film, similarly to radiographic film (see Sect. 17.4.3), has several features of importance to radiation dosimetry, such as high spatial resolution, permanent storage of processed image allowing multiple non-destructive readouts, and convenient physical shape and flexibility of the dosimeter cavity. In addition as discussed in Sect. 17.4.5, radiochromic film, in comparison with radiographic film, offers several important advantages, such as better water equivalence, insensitivity to visible light, and self-development after exposure.

17.5 Relative Radiation Dosimetry: Summary

Radiation dosimetry systems comprising a radiation sensitive dosimeter (cavity) and a reader are divided into two major categories: absolute and relative. An absolute dosimetry system is defined as a system whose response can be used to infer directly the dose absorbed in the dosimeter cavity without any need for calibration of the dosimeter signal in a known radiation field. A relative dosimetry system, on the other hand, is a system that requires calibration of its response in a known radiation field.

Three absolute dosimetry systems are known: calorimetric, chemical (Fricke), and ionometric and they were discussed in detail in Chap. 16. In contrast, a large variety of relative dosimetry systems are known ranging from very useful and practical, through mundane, all the way to impractical and of only academic interest. This chapter deals with relative dosimetry systems that are of practical interest to medical physics and are used in radiation medicine for calibration of high technology equipment used for diagnostic imaging or treatment of disease with ionizing radiation.

Four categories of relative dosimetry are used in radiation medicine: ionographic, luminescence, semiconductor, and film. Each of these broad categories is split into several subcategories and each of the subcategories can be classified as either and

Table 17.5 Four categories of relative radiation dosimetry systems discussed in this chapter

	Category of dosimetry system	Dosimeter	Mode
1.	Relative Ionometric Dosimetry	Parallel-plate ionization chamber	A
		Thimble (cylindrical) ionization chamber	A
		Area survey meter (ionization chamber)	A
		Re-entrant well ionization chamber	A
2.	Luminescence Dosimetry	Thermoluminescence (TL)	P
		Optically stimulated luminescence (OSL)	P
3.	Semiconductor Dosimetry	Diode (p-n junction)	A
		Diamond dosimeter	A
4.	Film Dosimetry	Radiographic film	P
		Radiochromic film	P

For each category the dosimeters discussed in this chapter as well as their mode of operation (active A or passive P) are also presented